

ECO362 Midterm Solutions

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October 26, 2023

Q1

(a) We solve for $y_b(2)$:

$$\begin{aligned}\exp(y_f(1,2)) &= \frac{\exp(2y_b(2))}{\exp(y_b(1))} \\ y_f(1,2) &= 2y_b(2) - y_b(1) \\ y_b(2) &= \frac{1}{2}(y_f(1,2) + y_b(1)) = \frac{1}{2}(0.42\% + 1.10\%) = 0.76\%\end{aligned}$$

- 5pt: Correct
- 3pt: Correct setup; incorrect calculations
- 2pt: Correct answer with little work

(b) The 3-year par yield is the coupon rate at which a 3-year coupon bond would trade at face value. Assuming no arbitrage, we solve for the 3-year par yield:

$$\begin{aligned}c(3) &= \frac{1 - P_b(3)}{\sum_{t=1}^3 P_b(t)} = \frac{1 - P_b(3)}{P_b(1) + P_b(2) + P_b(3)} \\ c(3) &= \frac{1 - \frac{1}{\exp(3y_b(3))}}{\frac{1}{\exp(y_b(1))} + \frac{1}{\exp(2y_b(2))} + \frac{1}{\exp(3y_b(3))}} \\ c(3) &= \frac{1 - \frac{1}{\exp(0.0296)}}{\frac{1}{\exp(0.0042)} + \frac{1}{\exp(0.0152)} + \frac{1}{\exp(0.0296)}} \\ c(3) &\approx \frac{0.029166}{0.99581 + 0.98491 + 0.97083} \approx 0.009882\end{aligned}$$

Note that $3y_b(3) = 2.96\%$ by the same type of calculation as in part (a), so we can numerically solve for $c(3)$ and obtain $c(3) \approx 0.9882\%$.

- 5pt: Correct
- 3pt: Correct setup; incorrect calculation but thorough attempt where $P_b(3)$ was calculated in the right manner

- 1pt: Correct setup with little calculation attempt or calculation attempt revealing conceptual misunderstanding

(c) Under no arbitrage, we can price the annuity using zero coupon bond prices:

$$P_a(2) = 100(P_b(1) + P_b(2))$$

$$P_a(2) = 100\left(\frac{1}{\exp(y_b(1))} + \frac{1}{\exp(2y_b(2))}\right)$$

$$P_a(2) = 100\left(\frac{1}{\exp(0.0042)} + \frac{1}{\exp(0.0152)}\right) \approx 100(0.9958 + 0.9849) \approx 198.07$$

- 5pt: Correct
- 3pt: Correctly specify the equation and thorough calculation attempt
- 1pt: Correct formula with little or no attempted calculation

(d) We would like to calculate $y_f(2, 5)$:

$$\exp(3y_f(2, 5)) = \frac{\exp(5y_b(5))}{\exp(2y_b(2))}$$

Note that $5y_b(5) = 0.42\% + 1.10\% + 1.44\% + 1.66\% + 1.75\% = 6.37\%$, so:

$$3y_f(2, 5) = 5y_b(5) - 2y_b(2) = 6.37\% - 1.52\% = 4.85\% \implies y_f(2, 5) \approx 1.617\%$$

- 5pt: Correct
- 3pt: Correct setup; incorrect calculations
- 2pt: Correct answer; no work shown

(e) Applying continuous compounding to the initial value of \$10,000:

$$\$10,000 \times \exp(6.37\%) \approx \$10,657.73$$

- 5pt: Correct
- 3pt: Correct setup; incorrect calculation
- 2pt: Correct setup but used annual compounding

Q2

- (a) Suppose the price of a security that pays \$1 in expansion and \$0 in recession is m_1 , and that of a security that pays \$0 in expansion and \$1 in recession is m_2 . Then by no arbitrage theory, and using the the price information of Amazon and Nvidia, we should have

$$\begin{cases} 164m_1 + 113m_2 = 126, \\ 619m_1 + 354m_2 = 442 \end{cases} \implies \begin{cases} m_1 = 0.4493, \\ m_2 = 0.4630. \end{cases} \quad (1)$$

- 5pt: Correctly specify the equations for which the prices of the securities must satisfy and correct numerical answer.
 - 3pt: Correctly specify the equations for which the prices of the securities must satisfy.
 - 2pt: Correct numerical answer.
- (b) See part (a).
- (c) To replicate \$1 of riskless cashflow in year 1 (in both states), one must buy one unit of security in (a) and one unit of security in (b). This portfolio costs $(m_1 + m_2)$, so the 1-year annually compounded riskless interest rate r must satisfy

$$m_1 + m_2 = \frac{1}{1+r}. \quad (2)$$

which implies $r = 9.62\%$.

- 5pt: Correct equation for which the riskless interest rate must satisfy and correct numerical answer.
 - 3pt: Correct equation.
 - 2pt: Correct numerical answer.
- (d) The future price of Apple in expansion p must satisfy

$$p \times m_1 + 138m_2 = 172, \quad (3)$$

which implies $p = 240.6260$.

- 5pt: Correct equation for which the the future price of Apple must satisfy and correct numerical answer.
 - 3pt: Correct equation.
 - 2pt: Correct numerical answer.
- (e) The future price of Tesla in recession p must satisfy

$$210m_1 + p \times m_2 = 247, \quad (4)$$

which implies $p = 329.6871$.

- 5pt: Correct equation for which the the future price of Tesla must satisfy and correct numerical answer.
- 3pt: Correct equation.
- 2pt: Correct numerical answer.

Q3

(a) Your wealth at the end of 2008 is $\$1,000(1 + 0.154)(1 + 0.057)(1 - 0.3674) = \771.63

- 5pt: Correctly specify the equation and correct numerical answer.
- 3pt: Correctly specify the equation but wrong answer with calculation error.
- 2pt: Try to specify but some errors in the equation and answer.
- 2pt: Correct numerical answer.

(b) Optimal weight $w^* = \frac{E(r_s) - r_b}{\gamma \sigma^2}$. $\gamma = \frac{0.12 - 0.048}{0.6 \times 0.2^2} = 3.00$

- 5pt: Correctly specify the equation and correct numerical answer.
- 3pt: Correctly specify the equation but wrong answer calculation error.
- 2pt: Try to specify but some errors in the equation and answer.
- 2pt: Correct numerical answer.

(c) When rebalancing, your portfolio return in each year will be $r_{p,t} = w r_{s,t} + (1 - w) r_{b,t}$, which yields 11.16%, 5.28%, and -21.40% from 2006 to 2008, respectively. Then, the wealth in your portfolio at the end of 2008 will be $\$1,000(1 + 0.1116)(1 + 0.0528)(1 - 0.2140) = \919.84 .

- 5pt: Correctly specify the equations and correct numerical answer.
- 3pt: Correctly specify the equations about the portfolio return ($r_{p,t}$) and cumulative return but but wrong answers with calculation error.
- 2pt: Right attempt to calculate the portfolio return but errors in equations and answers.
- 2pt: Correct numerical answer
- 1pt: Calculate cumulative returns without rebalancing.

(d) Unless your gamma is very small, according to the mean-variance theory it is always optimal to invest some portion of your wealth in the riskless asset. By doing so, you can have a better risk-return payoff. In this example, the stock market experienced a huge loss in 2008 after recording a higher return than bond for two years. If you put all your wealth in stock, then you would experience greater loss since your position in stock was larger.

- 5pt: Correct reasoning and explanation
- 3pt: Discussion of the mean-variance theory or optimal portfolio weights but with some incorrect explanation
- 3pt: Only mentioning the diversification effect or the fact that stocks are risky assets (not discussing the different level of risk-aversion of heterogeneous investors. The question asked why it is not optimal to some investors. A stock-only portfolio can be optimal depending on your level of risk-aversion.)
- 2pt: Briefly mentioning risk aversion or rebalancing but without enough explanation (e.g., not relating it to answers in (a) and (c))
- 2pt: Only mentioning historical stock market crash

Q4

(a)

Theory tells us that the portfolio that achieves the highest Sharpe ratio is the tangency portfolio. Its weights are given by:

$$w = \frac{\Sigma^{-1} (\mu - r_b \tilde{1})}{\gamma}$$

with

$$\Sigma = \begin{bmatrix} \sigma_{US}^2 & \text{Cov}(R_{US}, R_{CA}) \\ \text{Cov}(R_{US}, R_{CA}) & \sigma_{CA}^2 \end{bmatrix} = \begin{bmatrix} 4.3^2 & 0 \\ 0 & 5.65^2 \end{bmatrix} \times 10^{-4}$$

$$\mu = \begin{bmatrix} \mu_{US} \\ \mu_{CA} \end{bmatrix} = \begin{bmatrix} 0.87 \\ 0.89 \end{bmatrix} \times 10^{-2}$$

$$r_b \tilde{1} = r_b \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.08 \\ 0.08 \end{bmatrix} \times 10^{-2}$$

The inverse of Σ is

$$\Sigma^{-1} = \begin{bmatrix} \sigma_{US}^{-2} & 0 \\ 0 & \sigma_{CA}^{-2} \end{bmatrix} = \begin{bmatrix} 4.3^{-2} & 0 \\ 0 & 5.65^{-2} \end{bmatrix} \times 10^4$$

Then, substituting gives us the optimal solution:

$$\begin{bmatrix} w_{US}(10) \\ w_{CA}(10) \end{bmatrix} = \begin{bmatrix} (\mu_{US} - r_b) \cdot \sigma_{US}^{-2} \\ (\mu_{CA} - r_b) \cdot \sigma_{CA}^{-2} \end{bmatrix} \times \frac{1}{10} \approx \begin{bmatrix} 0.43 \\ 0.25 \end{bmatrix}$$

which leaves a share of $1 - 0.43 - 0.25 = 0.32$ for the riskless asset.

- 6 points: correct;
- 5 points: omit calculation of the riskless asset
- 4 points: correct equations but calculation error / incorrect matrix inversion;
- 3 points: minor incorrect specification like not dividing by the proper γ
- 0 points: otherwise

(b)

The expected excess return of the portfolio is the weighted average of the excess returns:

$$\mu_p - r_b = (0.43 \cdot 0.79 + 0.25 \cdot 0.81) \times 10^{-2} = 0.54\%$$

While the variance of the portfolio:

$$\sigma_p^2 = 0.43^2 \times \sigma_{US}^2 + 0.25^2 \times \sigma_{CA}^2 = 5.41 \times 10^{-4}$$

, hence the standard deviation is $\sigma_p = 2.33\%$ Therefore the Sharpe ratio is:

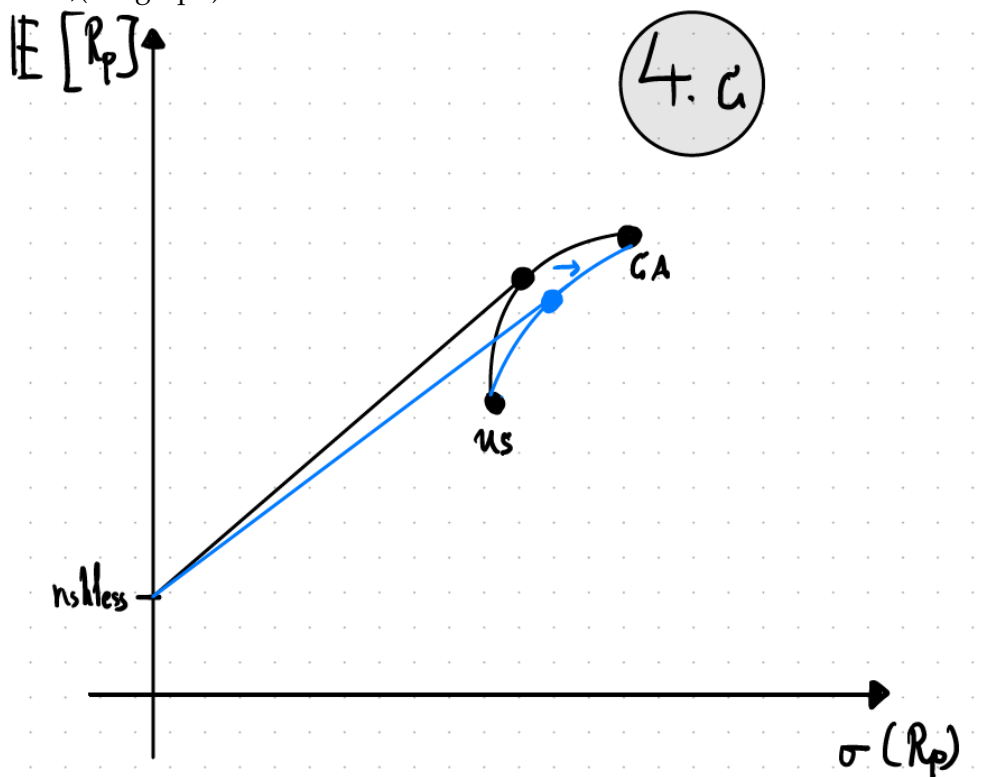
$$SR_p = \frac{\mu_p - r_b}{\sigma_p} \approx \frac{0.54\%}{2.33\%} = 0.23$$

- 6points: correct; numerical carry-over mistake
- 4 points: correct setup; incorrect calculation
- 3 points: mistake in standard deviation specification
- 2 points: correct setup;

(c)

By the separation theorem the Sharpe ratio of the optimal portfolio is the same as the Sharpe ratio of the tangency portfolio. Given the efficient frontier curve, the tangency portfolio is characterized as the tangency point of the tangent to the curve that intersects the y-axis at the riskless rate; the slope of the line is equal to the Sharpe ratio.

Due to the higher correlation, for any mixture of the two risky assets the associated risk will be higher. This is equivalent to shifting the efficient frontier to the right. Since the riskless rate remains unchanged the new tangency portfolio will have a **lower** Sharpe ratio, so will the optimal portfolio;(see graph)

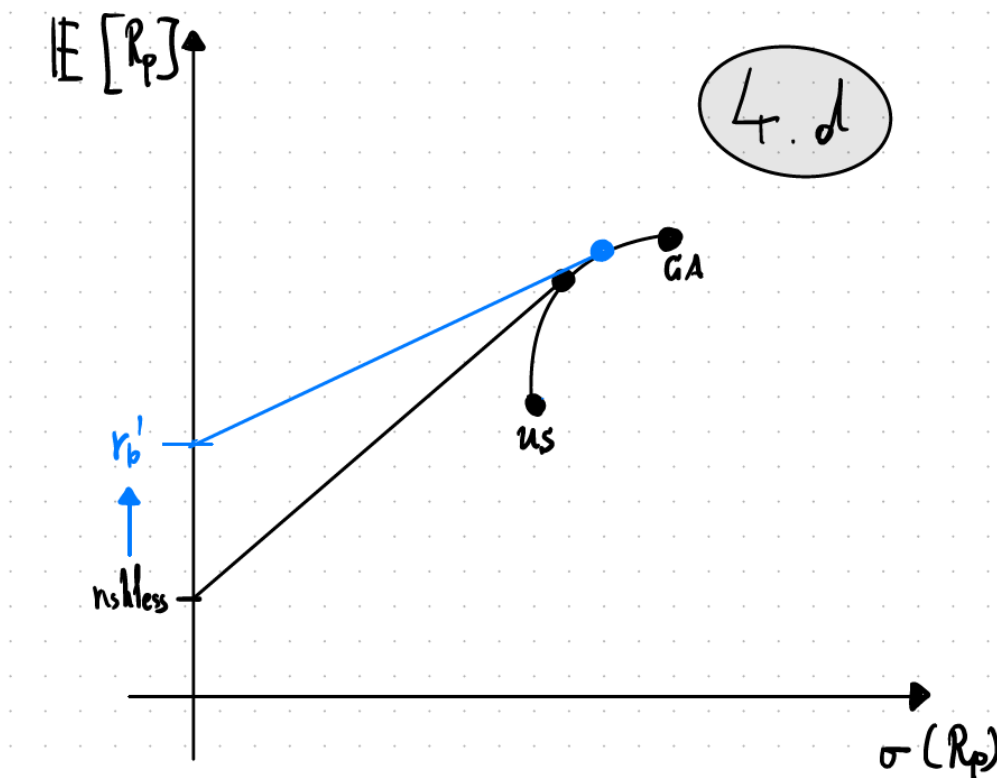


- 6points: correct

- 4 points: correct graph but false statement in explanation/ small mistake in graph (such as new CAL passing for new y-intercept)/ correct graph but for $\rho=1$
- 3 points: simply mention increased variance due to less diversification without mentioning variation of efficient frontier and tangency portfolio
- 2 points: Not link higher variance with less diversification / multiple mistakes in graph but correct answer / extra false statement (such as tangency portfolio or numerator of SR stays unchanged)
- 1 point: two false statements or not link higher variance with less diversification and extra false statement / correctly graph new efficient frontier with incorrect answer
- 0 points: otherwise

(d)

Since the efficiency frontier remains constant, an increase in the interest rate will induce the line tangent to the efficient frontier to become flatter. Hence, the new Sharpe ratio will be **lower** than part (b) (smaller Sharpe Ratio); see graph.



The careful student would note that there was a typo, the new interest rate should have been 0.1% but it was 1%. This is higher than the expected returns of both assets; therefore, the optimal portfolio for all agents is to simply hold the riskless asset; a strategy that does not have a well-defined Sharpe Ratio

- 6 points: correct

- 2 points: simply mention decreased excess ratio without recognizing variation of tangency portfolio/ keep tangency portfolio invariant on graph
- 1 point: correct final answer but including false statements, such as claim that the variance of the tangency portfolio is unchanged/ correct graph but incorrect answer
- 0 points: otherwise

(e)

Denote by w_{15}, w_{10} the optimal portfolio weight vectors on risky assets for agents with risk aversion $\gamma = 15$ and $\gamma = 10$ respectively.

Denote by $\hat{\gamma}$ the risk-aversion of an investor that exclusively invests on risky assets, which is the tangency portfolio, w^* . Then, by the separation theorem each investor puts a fraction $\frac{\hat{\gamma}}{\gamma}$ on the tangency portfolio.

Then,

$$w_{15} = \frac{\hat{\gamma}}{15} \cdot w^* = \frac{10}{15} \cdot \frac{\hat{\gamma}}{10} \cdot w^* = \frac{2}{3} \cdot w_{10} \Rightarrow \begin{bmatrix} w_{US}(15) \\ w_{CA}(15) \end{bmatrix} = \begin{bmatrix} 0.29 \\ 0.17 \end{bmatrix}$$

which leaves $w_b(15) = 0.54$

- 6 points: correct; numerical carry-over mistake
- 5 points: omit weight on riskless asset
- 4 points: correct set-up/ incorrect calculation
- 3 points: redundant calculation of tangency portfolio(if not already calculated in part 4.a) followed by application of separation theorem
- 1 point: apply separation theorem for the riskless asset
- 0 points: incorrect or no usage of separation theorem(for example use formula to calculate weights as in 4.a)